

Mahmoud Selim, PhD

Associate Professor

Faculty of Engineering, Alamein International University (AIU), Egypt

PUBLIC CONTACT

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Web of Science ResearcherID: AAA-3231-2021

ACADEMIC PROFILE

Researcher and educator working on intelligent wireless systems for smart cities and next-generation networks. Research interests include integrated sensing and communications (ISAC), UAV-assisted and RIS-assisted communications, Internet of Things (IoT), wireless sensor networks, physical-layer security, non-orthogonal multiple access (NOMA), B5G/6G enabling technologies, signal processing, and radio resource allocation.

ACADEMIC APPOINTMENTS

Associate Professor Faculty of Engineering, Alamein International University (AIU), Egypt	Nov. 2024 - Present
Associate Professor (tenured; on leave) Faculty of Engineering, Tanta University, Egypt	Jun. 2021 - Present
Assistant Professor Faculty of Engineering, Tanta University, Egypt	Oct. 2015 - Jun. 2021

RESEARCH EXPERIENCE

Postdoctoral Research Fellow Department of Information Engineering, University of Padova, Italy	Oct. 2022 - Oct. 2023
Postdoctoral Research Fellow College of Electronics and Information Engineering, Shenzhen University, China	Jul. 2018 - Jul. 2020
Special Research Student Graduate School of Information Science and Electrical Engineering, Kyushu University, Japan Supervisors: Osamu Muta, PhD and Hiroshi Furukawa, PhD.	Apr. 2014 - Jan. 2015

EDUCATION

PhD, Electronics and Communications Engineering Egypt-Japan University of Science and Technology (E-JUST), Alexandria, Egypt Thesis: <i>Radio Resource Management in OFDMA Wireless Systems</i> . Advisors: Hossam Shalaby, PhD and Said Elnoubi, PhD. Japanese co-advisors: Osamu Muta, PhD and Hiroshi Furukawa, PhD.	2015
MSc, Electronics and Communications Engineering Egypt-Japan University of Science and Technology (E-JUST), Alexandria, Egypt Thesis: <i>Enhanced Techniques for Interference Management in LTE Femtocell Networks</i> . Advisors: Mohamed El-Sharkawy, PhD, Hossam Shalaby, PhD, and Mostafa El-Khamy, PhD.	2012
BSc, Electronics and Electrical Communications Faculty of Engineering, Tanta University, Egypt Summa Cum Laude. Graduation project: <i>Enhancement and Compression of Medical Images using FPGA</i> .	2008

RESEARCH AND TECHNICAL SKILLS

- Programming: C, C++, MATLAB, and Python, including Keras and TensorFlow.
- Optimization and technical tools: CVX, YALMIP, Mathematica, LaTeX, and Microsoft Office.
- Operating systems: Microsoft Windows, Linux, and macOS.
- English proficiency: TOEFL iBT score 93.

REFEREED JOURNAL PUBLICATIONS

- Mohamed A. Aboulhassan, Ali G. Mohamed, and Mahmoud M. Selim. "Diversity Collapse in Dual-Hop UWOC Systems: The Impact of UAV Angular Misalignment." *IEEE Wireless Communications Letters*, vol. 15, Aug. 2026.
- Mahmoud M. Selim, Mohamed Rihan, Armin Dekorsy, and M. A. Aboulhassan. "Dynamic Role Allocation in Cooperative Secure UAV-ISAC Networks." *IEEE Open Journal of the Communications Society*, vol. 7, Aug. 2026.
- Mahmoud M. Selim. "Stochastic Geometry Analysis of UAV-assisted Networks with Probabilistic UAV Activation." *Scientific Reports*, vol. 15, no. 1, Oct. 2025.
- Aml Fawzy, Mahmoud M. Selim, Maha Elsabrouty, Sameh Napoleon, and Mustafa M. Abd El Naby. "Performance Analysis of Secure NOMA Wireless System over Rician Fading Channels." *Discover Applied Sciences*, vol. 7, Jul. 2025.
- Aml Fawzy, Mahmoud M. Selim, Maha Elsabrouty, Sameh Napoleon, and Mustafa M. Abd El Naby. "Non-Orthogonal Multiple Access Empowered Physical Layer Security Systems: Review, Issues and Challenges." *Journal of Engineering Research*, vol. 9, Mar. 2025.
- Gehad I. Mohammed, Mahmoud M. Selim, and Mahmoud A. Attia. "Trajectory Optimization for UAV-assisted Energy Harvesting for Internet-of-Things." *Journal of Engineering Research*, vol. 9, Mar. 2025.

7. Mahmoud M. Selim and Stefano Tomasin. **"Performance Analysis of RIS-assisted Downlink NOMA Wireless Systems Under D2D Interference."** Digital Signal Processing, vol. 144, Oct. 2023.
8. Doaa S. A. Elsersty, Mahmoud M. Selim, and Amira S. Ashour. **"Deep Learning-Based Polyp Detection in Wireless Capsule Endoscopy Images."** Journal of Engineering Research, vol. 5, no. 4, pp. 61-66, Dec. 2021.
9. Amr H. Hussein, Nourhan M. Salem, Mohamed E. Nasr, and Mahmoud M. Selim. **"BER and Capacity/Spectral Efficiency Enhancement of MIMO Systems using Digital Antenna Arrays Beamforming."** American Scientific Research Journal for Engineering, Technology, and Sciences, vol. 76, no. 1, pp. 218-238, Feb. 2021.
10. Mahmoud M. Selim, Mohamed Rihan, Yatao Yang, and Junxian Ma. **"Optimal Task Partitioning, Bit Allocation and Trajectory for D2D-Assisted UAV-MEC Systems."** Peer-to-Peer Networking and Applications, vol. 14, pp. 215-224, Jan. 2021.
11. Sarah Salah, Mahmoud M. Selim, Hussein Seleem, and Mahmoud A. A. Ali. **"Reduced OoB Emission and PAPR Using Partial-OSLM Technique in 5G UPMC Systems."** Journal of Engineering Research, vol. 4, pp. 1-6, Dec. 2020.
12. Hossam M. Kasem, Mahmoud M. Selim, Ehab Mahmoud Mohamed, and Amr H. Hussein. **"DRCS-SR: Deep Robust Compressed Sensing for Single Image Super-Resolution."** IEEE Access, vol. 8, pp. 170618-170634, Sept. 2020.
13. Mohamed Rihan, Mohamed Elwekeil, Yatao Yang, Lei Huang, Chen Xu, and Mahmoud M. Selim. **"Deep-VFog: When Artificial Intelligence Meets Fog Computing in V2X."** IEEE Systems Journal, pp. 1-14, Aug. 2020.
14. Amr H. Hussein, Alaa M. Enany, Mahmoud M. Selim, and Mohamed E. Nasr. **"BER Performance Enhancement of MIMO Systems Using Hybrid Detection Techniques Based on Sphere Decoding."** American Scientific Research Journal for Engineering, Technology, and Sciences, vol. 63, no. 1, pp. 7-21, Jan. 2020.
15. Mohamed Rihan, Mahmoud M. Selim, Chen Xu, and Lei Huang. **"D2D Communication Underlying UAV on Multiple Bands in Disaster Area: Stochastic Geometry Analysis."** IEEE Access, vol. 7, pp. 156646-156658, Oct. 2019.
16. Mahmoud M. Selim, Mohamed Rihan, Yatao Yang, Lei Huang, Zhi Quan, and Junxian Ma. **"On the Outage Probability and Power Control of D2D Underlying NOMA UAV-Assisted Networks."** IEEE Access, vol. 7, pp. 16525-16536, Feb. 2019.
17. Mahmoud M. Selim, Osamu Muta, Hossam M. H. Shalaby, and Hiroshi Furukawa. **"Chunk-based Resource Allocation in Multi-cell OFDMA Systems with Fairness Guarantee."** IEICE Communications Express, vol. 4, no. 2, pp. 31-37, Feb. 2015.
18. Mahmoud M. Selim, Osamu Muta, Hossam M. H. Shalaby, and Hiroshi Furukawa. **"Low Complexity QoS-aware Subcarrier Allocation for MISO-OFDMA Systems."** IEICE Communications Express, vol. 3, no. 8, pp. 235-240, Aug. 2014.

REFEREED CONFERENCE PUBLICATIONS

1. Eslam Ateia, Mahmoud M. Selim, and Mustafa M. Abd El Naby. **"Evaluation and Prediction of X-ray Tube Failure in Medical Imaging Systems."** 12th International Japan-Africa Conference on Electronics, Communications, and Computations (JAC-ECC), 2024.
2. Mahmoud M. Selim and Stefano Tomasin. **"Physical Layer Authentication With Simultaneous Reflecting and Sensing RIS."** IEEE 97th Vehicular Technology Conference (VTC2023-Spring), 2023.
3. Mohamed Rihan, Mahmoud M. Selim, Chen Xu, and Lei Huang. **"Flying Radar: Altitude Optimization, Clutter Mitigation and Spectrum Sharing With MIMO Communication System."** IEEE 11th Sensor Array and Multichannel Signal Processing Workshop (SAM), 2020.
4. Mahmoud M. Selim, Osamu Muta, Hossam Shalaby, and Hiroshi Furukawa. **"Chunk-based Resource Allocation in MISO-OFDMA Systems with Fairness Provision."** Third International Japan-Egypt Conference on Electronics, Communications and Computers (JEC-ECC), 2015.
5. Mahmoud M. Selim, Osamu Muta, Hossam Shalaby, and Hiroshi Furukawa. **"Fairness-aware Chunk-Based Resource Allocation in Multi-Cell OFDMA Networks."** IEEE International Conference on Connected Vehicles and Expo (ICCVE), 2014.
6. Mahmoud M. Selim, Said Elnoubi, Hossam Shalaby, Osamu Muta, and Hiroshi Furukawa. **"Reduced Complexity QoS-aware Resource Allocation Technique for MISO-OFDMA Systems."** International Conference on Computing, Networking and Communications (ICNC), 2014.
7. Mahmoud M. Selim, Mostafa El-Khamy, and Mohamed El-Sharkawy. **"Enhanced Frequency Reuse Schemes for Interference Management in LTE Femtocell Networks."** IEEE International Symposium on Wireless Communication Systems (ISWCS), 2012.

SELECTED PRESENTATIONS

- Oral presentation, "Physical Layer Authentication With Simultaneous Reflecting and Sensing RIS," IEEE VTC2023-Spring, Florence, Italy, Jun. 2023.
- Oral presentation, "Chunk-based Resource Allocation in MISO-OFDMA Systems with Fairness Provision," JEC-ECC 2015, Fukuoka, Japan, Mar. 2015.
- Oral presentation, "Fairness-aware Chunk-Based Resource Allocation in Multi-Cell OFDMA Networks," ICCVE 2014, Vienna, Austria, Nov. 2014.
- Poster presentation, "Enhanced Frequency Reuse Schemes for Interference Management in LTE Femtocell Networks," ISWCS 2012, Paris, France, Aug. 2012.

TEACHING EXPERIENCE

Faculty of Engineering, Alamein International University Associate Professor	Oct. 2024 - Present
Faculty of Engineering, Tanta University Assistant Professor / Associate Professor	Sept. 2015 - Sept. 2024
Faculty of Engineering, Horus University of Egypt Part-time teaching appointment	Sept. 2023 - Sept. 2024
Undergraduate courses	
- ELE231 Introduction to Embedded Systems (Fall 2024-2026)	
- ELE234 Logic Design (Fall 2024-2026; Spring 2025-2026)	

- MAT315 Numerical Analysis (Fall 2024-2026)
- MAT411 Complex and Special Functions and Numerical Analysis (Fall 2024-2026)
- ELE444 Satellite Communication Systems (Fall 2025-2026)
- ELE445 Selected Topics in Communication Systems (Spring and Summer 2026)
- ELE235 Industrial Networks (Spring 2025-2026)
- ELE238 Data Communication and Computer Networks (Spring 2025-2026)
- MEC471 Micro-controller and Embedded Systems (Spring 2025-2026)
- MAT112 Mathematics II (Summer 2025-2026)
- CSE131 Logic Design (Spring 2026)
- EEC4122 Satellite Communications (2015-2017, 2020-2021, 2023)
- EEC3213 Optical Communications (2016-2017)
- EEC4230 Mobile Communication Systems (2016-2017, 2021, 2024)
- EEC3143 Electrical Communications (2016-2017, 2022-2023)
- EEC1202 Electronics (2017)
- EEC2146 Electronic Circuits and Measurements (2017)
- EEC3215 Microprocessor Applications in Communications (2018)
- EEC130 Advanced Programming I (2016)
- EPE114 Electrical Measurements and Testing (2024)

Postgraduate courses

- EEC707 Selected Topics in Wireless Communications (2016)
- EEC707 Selected Topics in Communication Networks (2017)
- EEC507 Optical Fiber Networks (2016)
- EEC614 Adaptive Signal Processing (2021-2022, 2024)

SUPERVISION

- Supervised 5 MSc students and 1 PhD student affiliated with the Department of Electronics and Electrical Communications Engineering, Faculty of Engineering, Tanta University.
- Supervised 5 final-year undergraduate projects.

AWARDS AND FUNDING

- Five-year scholarship from the Missions Department, Egyptian Ministry of Higher Education, for MSc and PhD study at E-JUST (2010).
- Distinctive Graduates Award, Egyptian Engineering Syndicate, 2008 graduating class (2010).
- First in Row Award, Electronics and Electrical Communications Department, Tanta University (2009).
- University of Padova travel grant for IEEE VTC2023-Spring, Florence, Italy (2023).
- E-JUST Center at Kyushu University travel grants for ICCVE 2014, Vienna, and JEC-ECC 2015, Fukuoka.
- E-JUST travel grant for ISWCS 2012, Paris, France.
- Japan International Cooperation Agency short-visit travel award to Kyushu University (2011).

COMMUNITY SERVICE

- Volunteer, organizing committee, first International Japan-Egypt Conference on Electronics, Communications and Computers (JEC-ECC 2012).
- Volunteer, E-JUST Graduate Student Association.
- Volunteer, organizing committee, 25th National Radio Science Conference (NRSC 2008).

Public version - last updated September 2026. Personal address and telephone details intentionally omitted.